CS3236 Information Theory

AY 25/26 Sem 2 — github/omgeta

1. Introduction

Data Compression

Settings:

i. Lossless: recover data exactly (e.g. `.zip`)
   - L bits for N distinct messages: $L \geq \log_2 N$

ii. Lossy: allow distortion for smaller size (e.g. `.jpeg`)

Approaches:

i. Variable-length: shorter codewords for higher-probability sequences

ii. Fixed-length: encode "typical" sequences; small failure probability for low-probability sequences

Source Coding Theorem (informal):

i. Average lossless compression $\geq H$ (entropy)

Data Communication

Binary Symmetric Channel: bits flip indep. w.p. δ

$$\mathbb{P}[\text{bit flip}] = \delta, \quad \mathbb{P}[\text{no flip}] = 1-\delta, \quad \delta \in (0, \frac{1}{2})$$

Approaches:

i. Uncoded:
   - $\mathbb{P}[\text{correct}] = (1-\delta)^n$
   - $R = 1$

ii. Repetition Code (odd $n, 1$):
repeat bits n times, decide by majority vote
   - $\mathbb{P}[\text{correct}] = \sum_{i=0}^{(n-1)/2} \binom{n}{i} \delta^i (1-\delta)^{n-i}$
   - $R = \frac{1}{n}$

iii. Hamming Code $(7, 4)$: 3 parity bits for 4 data bits (detect ≤ 2 errors, correct 1)
   - $\mathbb{P}[\text{correct}] = (1-\delta)^7 + 7\delta(1-\delta)^6$
   - Rate $R = \frac{4}{7}$

Channel Coding Theorem (informal):

i. If $R < C$ (capacity), $\exists$ codes s.t. $\mathbb{P}[\text{error}] \to 0$

2. Measures

Information

Information from event A w.p. p: $\psi(A) = \log(\frac{1}{p})$

i. Uncertainty of/Info gained from A occuring

Entropy

Entropy of X over $\mathcal{X}$:

$$H(X) = \mathbb{E}_{X \sim P_X}\left[\log_2 \frac{1}{P_X(X)}\right]$$

$$H_2(p) = p \log_2 \frac{1}{p} + (1-p) \log_2 \frac{1}{1-p}$$

i. $H(X) \geq 0$
with equality iff X deterministic (Non-negativity)

ii. $H(X) \leq \log_2 |\mathcal{X}|$
with equality iff $X \sim \text{Unif}(\mathcal{X})$ (Upper bound)

iii. $H(f(X)) \leq H(X)$
with equality iff f is one-to-one (Function)

Conditional Entropy of X, Y:

$$H(Y|X) = \mathbb{E}_{(X,Y) \sim P_{XY}}\left[\log_2 \frac{1}{P_{Y|X}(Y|X)}\right]$$
$$= \sum_x P_X(x) H(Y|X=x)$$

1. $H(Y|X) \leq H(Y)$
with equality iff $X \perp Y$ (Conditioning $\downarrow$)
2. If Y depends on X only through $f(X)$,
$H(Y|X) = H(Y|f(X))$ (Info-Preserving)

Joint Entropy of X, Y:

$$H(X,Y) = \mathbb{E}_{(X,Y) \sim P_{XY}}\left[\log_2 \frac{1}{P_{XY}(X,Y)}\right]$$

1. $H(X,Y) = H(Y,X)$ (Symmetric)
2. $H(X,Y) = H(X) + H(Y|X)$
$H(X_1, \ldots, X_n) = \sum_{i=1}^n H(X_i|X_1, \ldots, X_{i-1})$(Chain)
   - $H(X,Y|Z) = H(X|Z) + H(Y|X,Z)$
3. $H(X_1, \ldots, X_n) \leq \sum_{i=1}^n H(X_i)$
with equality iff $X_1, \ldots, X_n$ indep. (Sub-additivity)

Relative Entropy (KL Divergence)

Relative entropy of PMFs P_X, Q_X:

$$D(P\|Q) = \mathbb{E}_{X \sim P}\left[\log_2 \frac{P(X)}{Q(X)}\right]$$

i. $D(P\|Q) \geq 0$
with equality iff $P = Q$ (Non-negativity)

ii. $D(P_{XY}\|Q_{XY}) = D(P_X\|Q_X) + D(P_{Y|X}\|Q_{Y|X}|P_X)$ (Chain)

iii. P_Y, Q_Y from applying $V(y|x)$ to P_X, Q_X
$\implies D(P_Y\|Q_Y) \leq D(P_X\|Q_X)$ (DPI)

Mutual Information

Mutual Information of X, Y over $\mathcal{X}, \mathcal{Y}$:

$$\begin{aligned} I(X;Y) &= H(Y) - H(Y|X) \\ &= H(X) + H(Y) - H(X,Y) \\ &= D(P_{XY}\|P_X \times P_Y) \end{aligned}$$

i. $I(X;Y) = I(Y;X)$ (Symmetric)

ii. $I(X;Y) \geq 0$
with equality iff $X \perp Y$ (Non-negativity)

iii. $I(X;Y) \leq H(X) \leq \log_2 |\mathcal{X}|$
$I(X;Y) \leq H(Y) \leq \log_2 |\mathcal{Y}|$ (Upper bounds)

Conditional Mutual Information for X, Y, Z:

$$I(X;Y|Z) = H(Y|Z) - H(Y|X,Z)$$

i. $I(X;Y|Z) \geq 0$
with equality iff $X \perp Y \mid Z$ (Non-negativity)

Joint Mutual Information for X, Y, Z:

$$\begin{aligned} I(X,Y;Z) &= H(Z) - H(Z|X,Y) \\ &= I(X;Z) + I(Y;Z|X) \\ I(X;Y,Z) &= I(X;Y) + I(X;Z|Y) \end{aligned}$$

i. $I(X_1, \ldots, X_n; Y) = \sum_{i=1}^n I(X_i; Y|X_1, \ldots, X_{i-1})$ (Chain)

ii. $X \to Y \to Z \Rightarrow I(X;Z) \leq I(X;Y)$
with equality iff $I(X;Y|Z) = 0$ (DPI)

iii. If $(Y_i \perp Y_j) \mid (X_1, \ldots, X_n) \wedge Y_i$ depends only on X_i,
$\implies I(X_1, \ldots, X_n; Y_1, \ldots, Y_n) \leq \sum_{i=1}^n I(X_i; Y_i)$ (Partial Sub-additivity)

3. Symbol-Wise Source Coding

Symbol-wise source coding maps each $x \in \mathcal{X}$ to some binary sequence $C(x)$ with length $\ell(x)$

$$L(C) = \mathbb{E}[\ell(X)] = \sum_{x \in \mathcal{X}} P_X(x)\ell(x)$$

Decodability Conditions:

i. Nonsingular: $x \neq x' \Rightarrow C(x) \neq C(x')$

ii. Uniquely Decodable:
$C(x_1) \cdots C(x_n)$ decodes to a unique $x_1, \ldots, x_n$
- $\sum_{x \in \mathcal{X}} 2^{-\ell(x)} \leq 1$ (Kraft's Inequality)

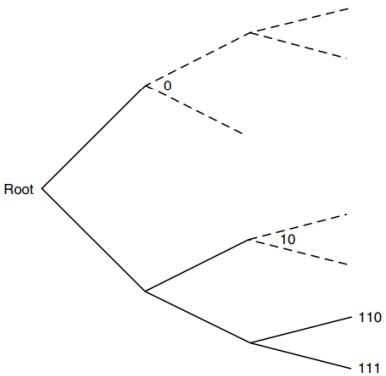
Prefix-free Code

Prefix-free codes have no codeword as prefix of another:

+ Uniquely decodable
+ Loses no optimality in average length
+ Instantaneous left-to-right decoding with greedy algorithm; termination uniquely identifies codeword

i. Existence: $\{\ell(x)\}_{x \in \mathcal{X}}$ satisfy Kraft's inequality $\iff \exists$ a prefix-free code mapping $x \in \mathcal{X}$ to codeword of length $\ell(x)$

ii. $L(C) \geq H(X)$
with equality iff $P_X(x) = 2^{-\ell(x)}$ (Entropy-Bound)
- Ideal length $\ell^*(x) = \log_2 \frac{1}{P_X(x)}$



Shannon–Fano Code

Shannon–Fano code rounds ideal lengths up to the nearest integer:

$$\ell(x) = \lceil \log_2 \frac{1}{P_X(x)} \rceil$$

i. $\sum_x 2^{-\ell(x)} \leq 1$ so $\exists$ prefix-free code with lengths

ii. $\log_2 \frac{1}{P_X(x)} \leq \ell(x) < \log_2 \frac{1}{P_X(x)} + 1$
$\implies H(X) \leq L(C) < H(X) + 1$

iii. Mismatch (true P_X but $\ell(x)$ uses Q_X):
$H(X) + D(P_X \| Q_X) \leq L(C) < H(X) + D(P_X \| Q_X) + 1$

Huffman Code

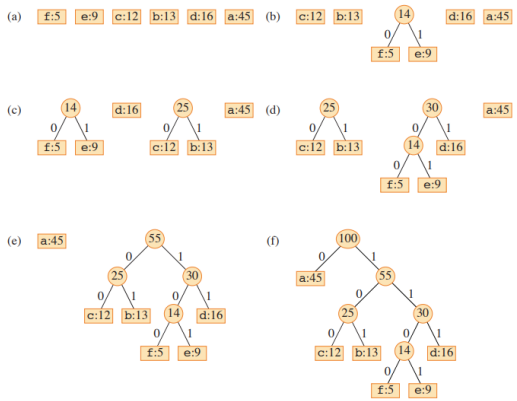
Huffman code is the optimal prefix-free symbol code minimizing $L(C)$:

i. Optimality:
for any prefix-free code C, $L(C_{\text{Huff}}) \leq L(C)$
- $H(X) \leq L(C_{\text{Huff}}) < H(X) + 1$
- $\sum_x 2^{-\ell(x)} = 1$ from optimality

Algorithm:

i. Construct by repeatedly merging the two least-probable nodes into one with probability sum

ii. Label merge edges 0/1; codeword is the root-to-leaf path for each symbol



Interpretation as Yes/No Questions

A sequence of yes/no questions that uniquely identifies $x \in \mathcal{X}$ is equivalent to a binary code:

i. Each symbol x has a distinct sequence of answers, hence a distinct binary codeword

ii. If symbol x is identified after $\ell(x)$ questions, then its codeword has length $\ell(x)$

iii. Conversely, a binary code induces a questioning scheme by asking for successive codeword bits

iv. Hence the average number of yes/no questions is

$$H(X) \leq \mathbb{E}[Q] = L(C) < H(X) + 1$$

v. Therefore, Huffman coding gives an optimal yes/no questioning scheme

Memory Limitation

Symbol-wise codes ignore dependence across symbols.

Mitigate via **block coding**:
group $\mathcal{X}^N$ and code w.r.t. $P_{X_1,\ldots,X_N}$:

+ Exploit sequence probabilities ("apple" $\gg$ "ealpp")
+ If i.i.d, $H(X_1, \ldots, X_N) = \sum_{i=1}^N H(X_i)$
$\Rightarrow NH(X) \leq L(C) \leq NH(X) + 1$
$\Rightarrow H(X) \leq$ Avg. length per symbol $\leq H(X) + \frac{1}{N}$

– Determining $P_{X_1,\ldots,X_N}$ accurately is hard
– Complexity grows exponentially: $|\mathcal{X}|^N$ symbols (sorting / Huffman)

Appendix

Random Variables

Probability Mass Function (PMF) for discrete X:

$$P_X(x) = \mathbb{P}[X = x]$$

i. $\forall x,\ 0 \le P_X(x) \le 1$
ii. $\sum_x P_X(x) = 1$

Probability Density Function (PDF) for continuous X:

$$\int_a^b f_X(x)\, dx = \mathbb{P}[a \le X \le b]$$

i. $\forall x,\ f_X(x) \ge 0$
ii. $\int_x f_X(x)\, dx = 1$

Expectation

Expectation of X:

$$\mathbb{E}[X] = \sum_x P_X(x)\, x$$

i. $\mathbb{E}[f(X)] = \sum_x P_X(x)\, f(x)$ (Function)
ii. $\mathbb{E}[c(X)] = c\mathbb{E}[X]$ (Scaling)
iii. $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$ (LoE)
iv. $\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y] \quad \text{if } X \perp Y$ (Product)
v. $\mathbb{E}[\mathbf{1}\{A\}] = \mathbb{P}[A]$ (Indicator)

Variance

Variance of X:

$$\begin{aligned}\mathrm{Var}[X] &= \mathbb{E}[(X-\mu)^2] \\ &= \mathbb{E}[X^2] - \mu^2\end{aligned}$$

i. $\mathrm{Var}[cX] = c^2\,\mathrm{Var}[X]$ (Scaling)
ii. $\mathrm{Var}[X+Y] = \mathrm{Var}[X] + \mathrm{Var}[Y] + 2\,\mathrm{Cov}[X,Y]$
$= \mathrm{Var}[X] + \mathrm{Var}[Y] \quad \text{if } X \perp Y$ (Sum)
iii. $\mathrm{Cov}[X,Y] = \mathbb{E}[(X-\mu_X)(Y-\mu_Y)]$ (Covariance)

Conditioning

Conditional:

$$P_{Y|X}(y|x) = \frac{P_{XY}(x,y)}{P_X(x)}$$

i. $\mathbb{P}[A] = \sum_x P_X(x)\,\mathbb{P}[A|X=x]$ (Total Probability)
ii. $\mathbb{E}[Y] = \mathbb{E}_X\big[\mathbb{E}_Y[Y|X]\big]$ (Total Expectation)
iii. $\mathbb{P}[A|B] = \frac{\mathbb{P}[A]\,\mathbb{P}[B|A]}{\mathbb{P}[B]}$ (Bayes)

Independence

Independence $X \perp Y$:

$$\begin{aligned}P_{XY}(x,y) &= P_X(x)\,P_Y(y) \\ P_{Y|X}(y|x) &= P_Y(y)\end{aligned}$$

i. $f(X) \perp g(Y)$ if f, g deterministic (Functions)
ii. Joint Indep.: $P_{X_1,\ldots,X_n}(x_1,\ldots,x_n) = \prod_{i=1}^n P_{X_i}(x_i)$
- Pairwise Indep. $\not\Rightarrow$ Joint Indep.

Conditional Independence $X \perp Y \mid Z$:

$$\begin{aligned}P_{XY|Z}(x,y|z) &= P_{X|Z}(x|z)\,P_{Y|Z}(y|z) \\ P_{X|YZ}(x|y,z) &= P_{X|Z}(x|z) \\ P_{Y|XZ}(y|x,z) &= P_{Y|Z}(y|z)\end{aligned}$$

Indicators

Indicator Random Variable:

$$\mathbf{1}\{A\} = \begin{cases} 1, & A \text{ occurs} \\ 0, & A \text{ does not occur}\end{cases}$$

i. $\mathbb{E}[\mathbf{1}\{A\}] = \mathbb{P}[A]$ (Probability)
ii. If $N = \sum_{i=1}^n \mathbf{1}\{A_i\}$, then $\mathbb{E}[N] = \sum_{i=1}^n \mathbb{P}[A_i]$ (Counting)
iii. Uses:
- Counting: count occurences of events
- Simplification: decompose complex RVs into simpler binary indicators

Other

Marginal Distributions:

$$P_X(x) = \sum_y P_{XY}(x,y), \quad P_Y(y) = \sum_x P_{XY}(x,y)$$

Union vs. Intersection:

$$\mathbb{P}[A \cup B] = \mathbb{P}[A] + \mathbb{P}[B] - \mathbb{P}[A \cap B]$$

Union Bound (Boole's Inequality):

$$\mathbb{P}\left[\bigcup_{i=1}^N A_i\right] \le \sum_{i=1}^N \mathbb{P}[A_i]$$

Law of Large Numbers for i.i.d. $X_1, \ldots, X_n$ with mean μ:

$$\mathbb{P}\left[\left|\frac{1}{n}\sum_{i=1}^n X_i - \mu\right| > \epsilon\right] \to 0 \quad \text{as } n \to \infty$$

Jensen's Inequality:

i. If f is convex, then $f(\mathbb{E}[X]) \le \mathbb{E}[f(X)]$
ii. If f is concave, then $f(\mathbb{E}[X]) \ge \mathbb{E}[f(X)]$

Ceiling Bound: $x \le \lceil x \rceil < x + 1$

Logarithm Properties:

i. $\log xy = \log x + \log y$
ii. $\log \frac{1}{x} = -\log x$
iii. $\log x^c = c \log x$
iv. $\log_a x = \frac{\log_b x}{\log_b a}$
v. $\log_e x \le x - 1$ with equality iff $x = 1$
vi. $0 \log \frac{0}{q} = 0$
vii. $p \log \frac{p}{0} = \infty$

Calculus Rules:

i. $\frac{d}{dx} x^c = c x^{c-1}$
ii. $\frac{d}{dx} e^{cx} = c e^{cx}$
iii. $\frac{d}{dx} \ln x = \frac{1}{x}$
iv. $\frac{d}{dx} f(g(x)) = f'(g(x)) g'(x)$ (Chain)
v. $\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$ (Product)
vi. $\frac{d}{dx}\frac{f(x)}{g(x)} = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$ (Quotient)